

CLINICAL FINDINGS

A consensus development conference was held by the National Institutes of Health in 1987 to establish diagnostic criteria for neurofibromatosis type 1 (NF-1), with the goal of improving clinical research and patient care. The conference identified seven diagnostic features (Table 142-1), and it was recommended that two or more of these criteria be present to establish a diagnosis of NF-1. These criteria have proven highly effective and remain in use without modification more than 20 years later.

Café-au-Lait Spots

Café-au-lait spots are flat, pigmented macules and are often the first clinical manifestation of NF-1 (Fig. 142-1). They are frequently present at birth and tend to increase in number during infancy and early childhood, with new spots commonly appearing throughout the first decade of life. As the child grows, existing spots also enlarge proportionally. Although they can occur anywhere on the body, they are rarely found on the face. The size, shape, and borders of café-au-lait spots are not diagnostically predictive; however, most spots in NF-1 are oval with smooth, well-defined margins.

The commonly cited notion that smooth-edged café-au-lait spots suggest NF-1 while irregular borders are more typical of McCune-Albright syndrome can be misleading and should be interpreted cautiously (Box 142-1). When café-au-lait spots overlap, the area of overlap may appear darker than either individual spot. When located within mongolian spots, they are often surrounded by a lighter pigmented halo.

The number or distribution of café-au-lait spots does not correlate with disease severity in NF-1. Additionally, the location of these macules does not predict where neurofibromas will develop.

Histologically, café-au-lait spots consist of clusters of heavily pigmented melanocytes of neural crest origin in the epidermis. These cells may contain

increased numbers of "giant" melanosomes or melanin macroglobules, but these findings are not specific to NF-1 and have no diagnostic utility in biopsy specimens.

TABLE 142-1 Diagnostic Criteria for Neurofibromatosis Type 1 (NF-1)^a

- Six or more café-au-lait macules >5 mm in greatest diameter in prepubertal individuals or >15 mm in postpubertal individuals
- Two or more neurofibromas of any type, or one plexiform neurofibroma
- Freckling in the axillary or inguinal regions
- Optic pathway glioma
- Two or more iris Lisch nodules
- A distinctive osseous lesion, such as sphenoid dysplasia or thinning of the long bone cortex, with or without pseudarthrosis
- A first-degree relative (parent, sibling, or offspring) with NF-1 by the above criteria

^a Two or more criteria required for diagnosis.

Box 142-1

Differential Diagnosis of Syndromes Associated with Multiple Café-au-Lait Spots^a

- Neurofibromatosis type 1
- Familial café-au-lait spots
- LEOPARD syndrome (multiple lentigines, electrocardiographic abnormalities, ocular hypertelorism, pulmonary stenosis, abnormalities of the genitalia, growth retardation, deafness)
- Neurofibromatosis type 2
- McCune-Albright syndrome (polyostotic fibrous dysplasia)
- Tuberous sclerosis
- Fanconi anemia

- Multiple endocrine neoplasia type 2B
- Bannayan-Riley-Ruvalcaba syndrome
- Bloom syndrome
- Ataxia-telangiectasia

^a Listed in approximate order of typical number of café-au-lait macules.